Causal Analysis Report

BPI_Challenge_2013_open_problems.xes.gz-flw

Decision points:	3170
Choice sets:	1
Transitions fitted:	5
Significant (p<0.05):	0
Generated:	2026-06-19 14:41

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

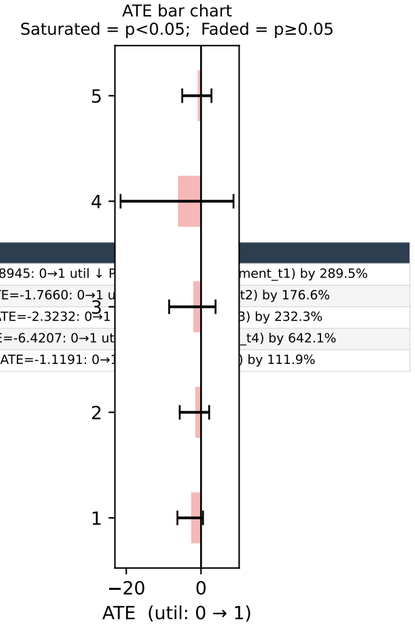
(no data)

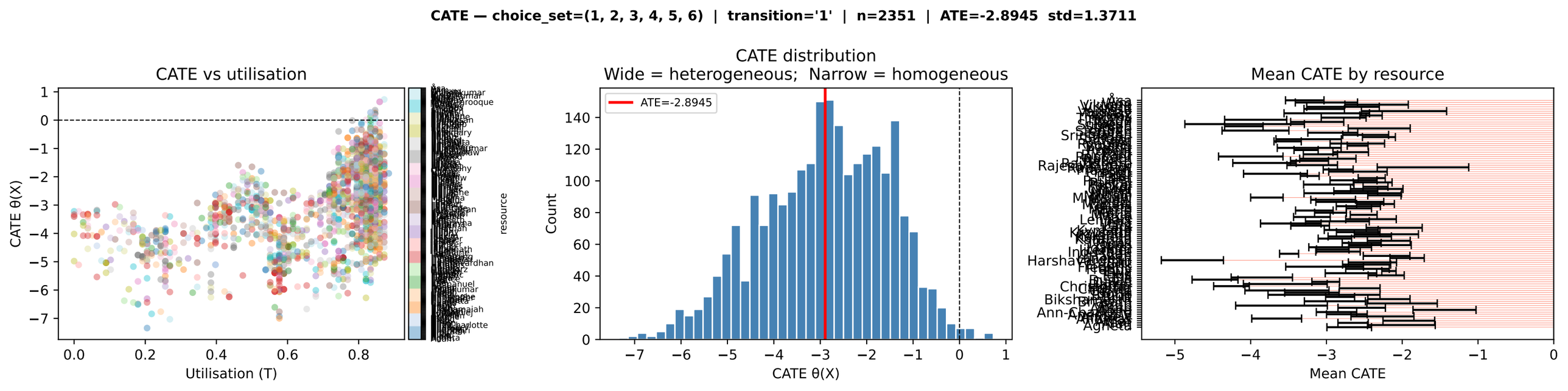
Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

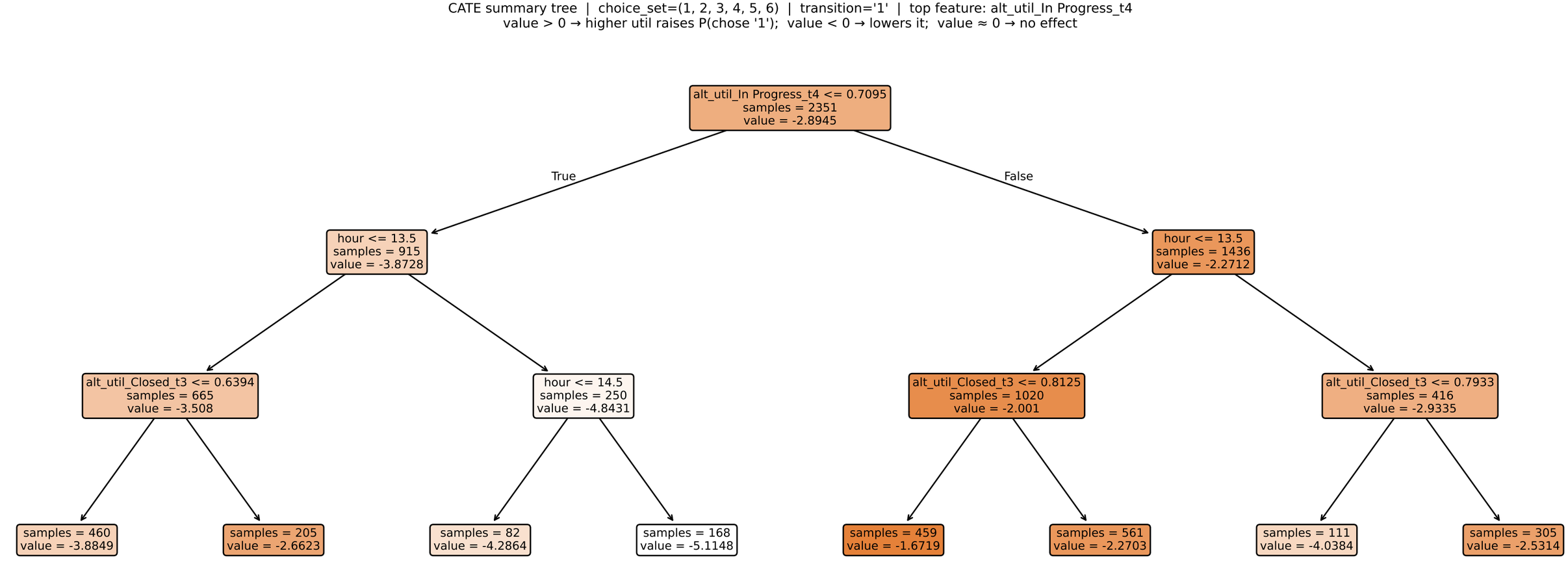
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	1	Awaiting Assignment_t1	Assigned_t2, Closed_t3, In Progress_t4, Wait_t5	-0.807745	0.246658	[-1.2912, -0.3243]	0.0011	True	θ=-0.8077: 1-unit ↑ util ↓ P(t1/Awaiting Assignment_t1) by 80.8%
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	2	Assigned_t2	Awaiting Assignment_t1, Closed_t3, In Progress_t4, Wait_t5	-0.014243	0.332523	[-0.6660, 0.6375]	0.9658	False	θ=-0.0142: 1-unit ↑ util ↓ P(t2/Assigned_t2) by 1.4%
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	3	Closed_t3	Assigned_t2, Awaiting Assignment_t1, In Progress_t4, Wait_t5	0.4133	0.22783	[-0.0332, 0.8598]	0.0697	False	θ=+0.4133: 1-unit ↑ util ↑ P(t3/Closed_t3) by 41.3%
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	4	In Progress_t4	Assigned_t2, Awaiting Assignment_t1, Closed_t3, Wait_t5	-3.077827	0.351853	[-3.7675, -2.3882]	0.0	True	θ=-3.0778: 1-unit ↑ util ↓ P(t4/In Progress_t4) by 307.8%
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	5	Wait_t5	Assigned_t2, Awaiting Assignment_t1, Closed_t3, In Progress_t4	0.28853	0.324642	[-0.3478, 0.9248]	0.3741	False	θ=+0.2885: 1-unit ↑ util ↑ P(t5/Wait_t5) by 28.9%

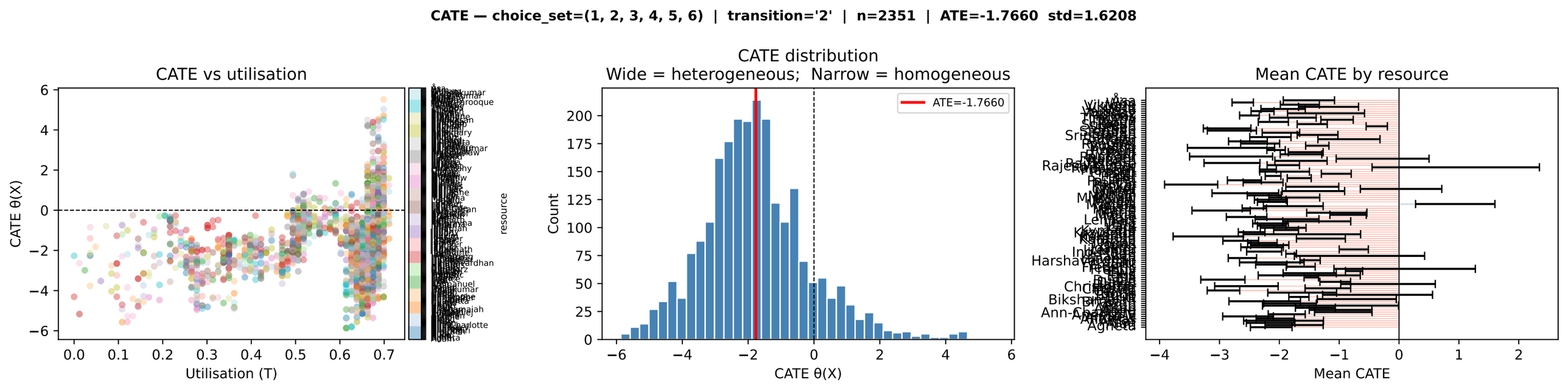
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

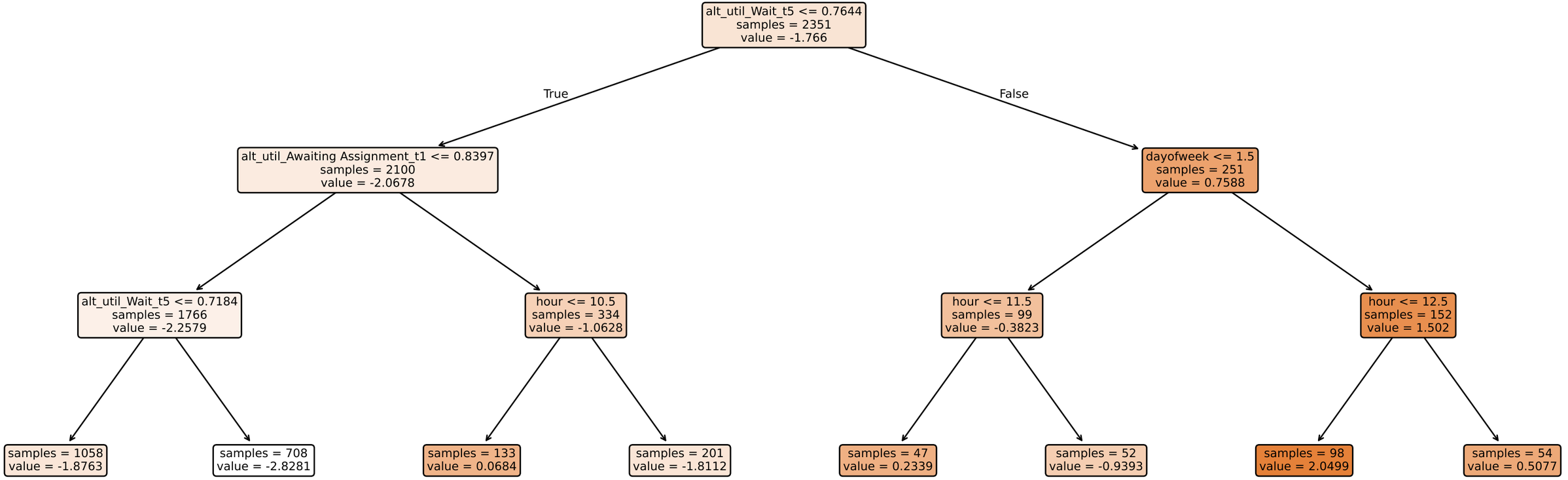
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	1	Awaiting Assignment_t1	Assigned_t2, Closed_t3, In Progress_t4, Wait_t5	-2.894506	[-6.3196, 0.5306]	0.0976	False	ATE=-2.8945: 0→1 util ↓ P…ment_t1) by 289.5%
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	2	Assigned_t2	Awaiting Assignment_t1, Closed_t3, In Progress_t4, Wait_t5	-1.766046	[-5.7048, 2.1727]	0.3795	False	ATE=-1.7660: 0→1 u…t2) by 176.6%
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	3	Closed_t3	Assigned_t2, Awaiting Assignment_t1, In Progress_t4, Wait_t5	-2.323152	[-8.5322, 3.8859]	0.4633	False	ATE=-2.3232: 0→1…3) by 232.3%
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	4	In Progress_t4	Assigned_t2, Awaiting Assignment_t1, Closed_t3, Wait_t5	-6.420711	[-21.5311, 8.6897]	0.4049	False	ATE=-6.4207: 0→1 ut…_t4) by 642.1%
(1, 2, 3, 4, 5, 6)	('Awaiting Assignment_t1', 'Assigned_t2', 'Closed_t3', 'In Progress_t4', 'Wait_t5', 'τ_t6')	2351	1, 2, 3, 4, 5	5	Wait_t5	Assigned_t2, Awaiting Assignment_t1, Closed_t3, In Progress_t4	-1.119095	[-5.0296, 2.7915]	0.5749	False	ATE=-1.1191: 0→1…) by 111.9%

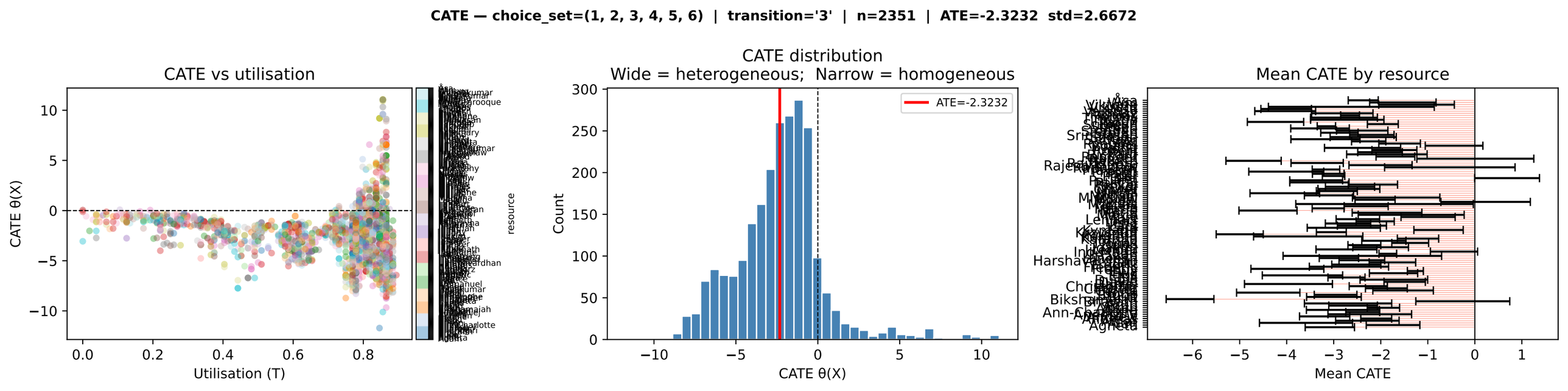


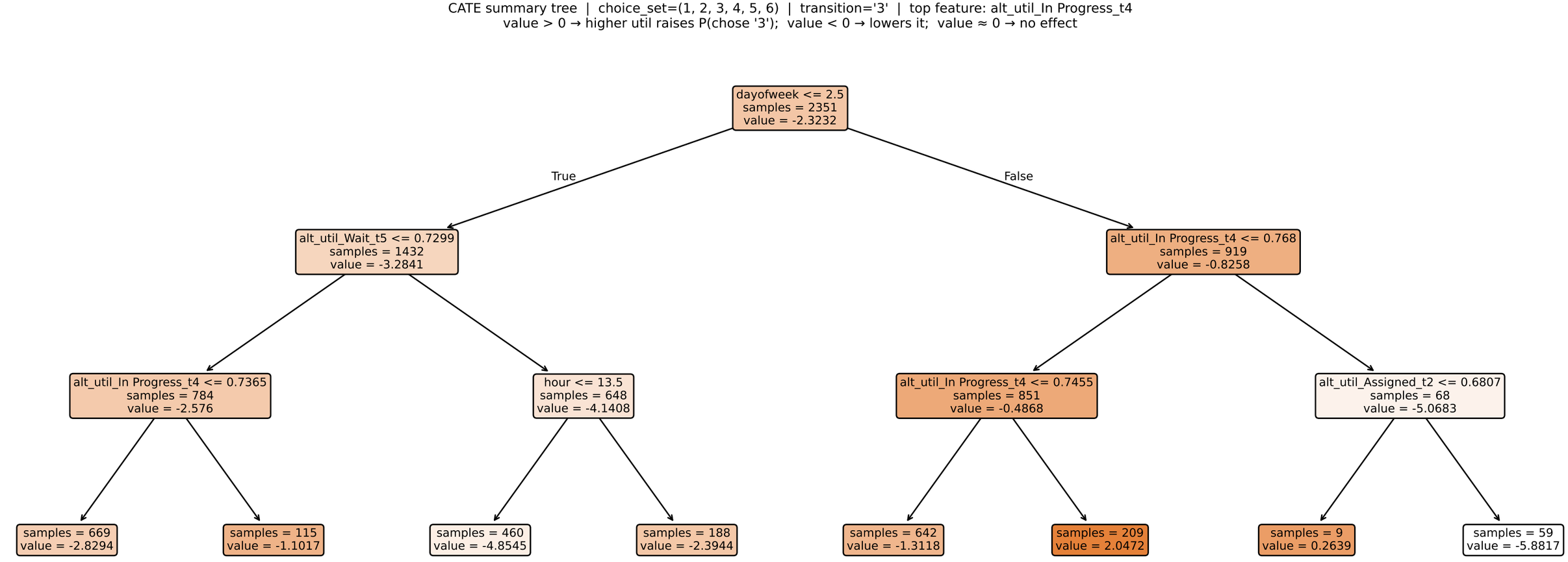

CATE — choice_set=(1, 2, 3, 4, 5, 6) | transition='1' | n=2351 | ATE=-2.8945 std=1.3711
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.8945
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

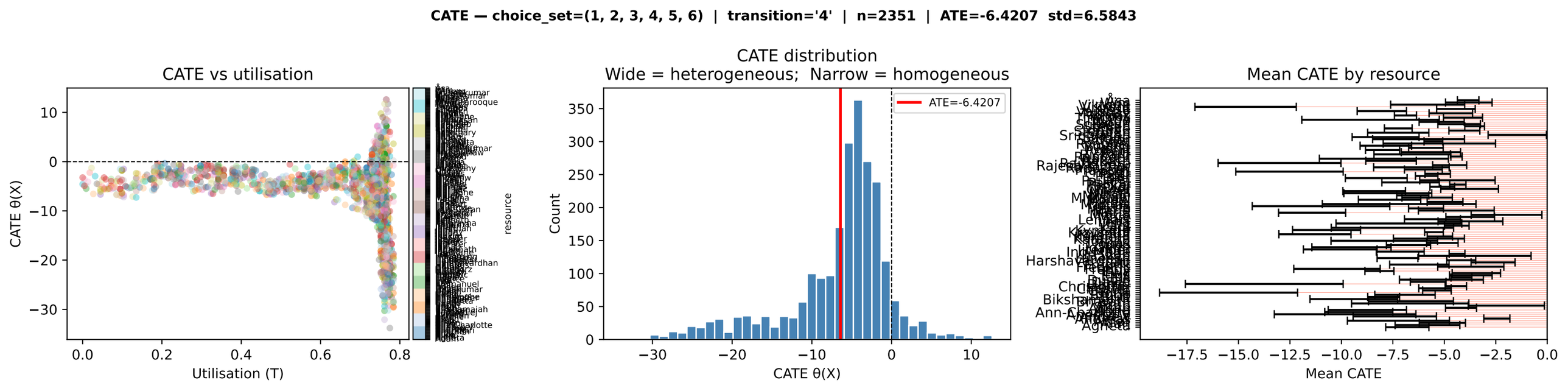

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6) | transition='1' | top feature: alt_util_ln Progress_t4
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_ln Progress_t4 <= 0.7095
samples = 2351
value = -2.8945
True
False
hour <= 13.5
samples = 915
value = -3.8728
hour <= 13.5
samples = 1436
value = -2.2712
alt_util_Closed_t3 <= 0.6394
samples = 665
value = -3.508
hour <= 14.5
samples = 250
value = -4.8431
alt_util_Closed_t3 <= 0.8125
samples = 1020
value = -2.001
alt_util_Closed_t3 <= 0.7933
samples = 416
value = -2.9335
samples = 460
value = -3.8849
samples = 205
value = -2.6623
samples = 82
value = -4.2864
samples = 168
value = -5.1148
samples = 459
value = -1.6719
samples = 561
value = -2.2703
samples = 111
value = -4.0384
samples = 305
value = -2.5314

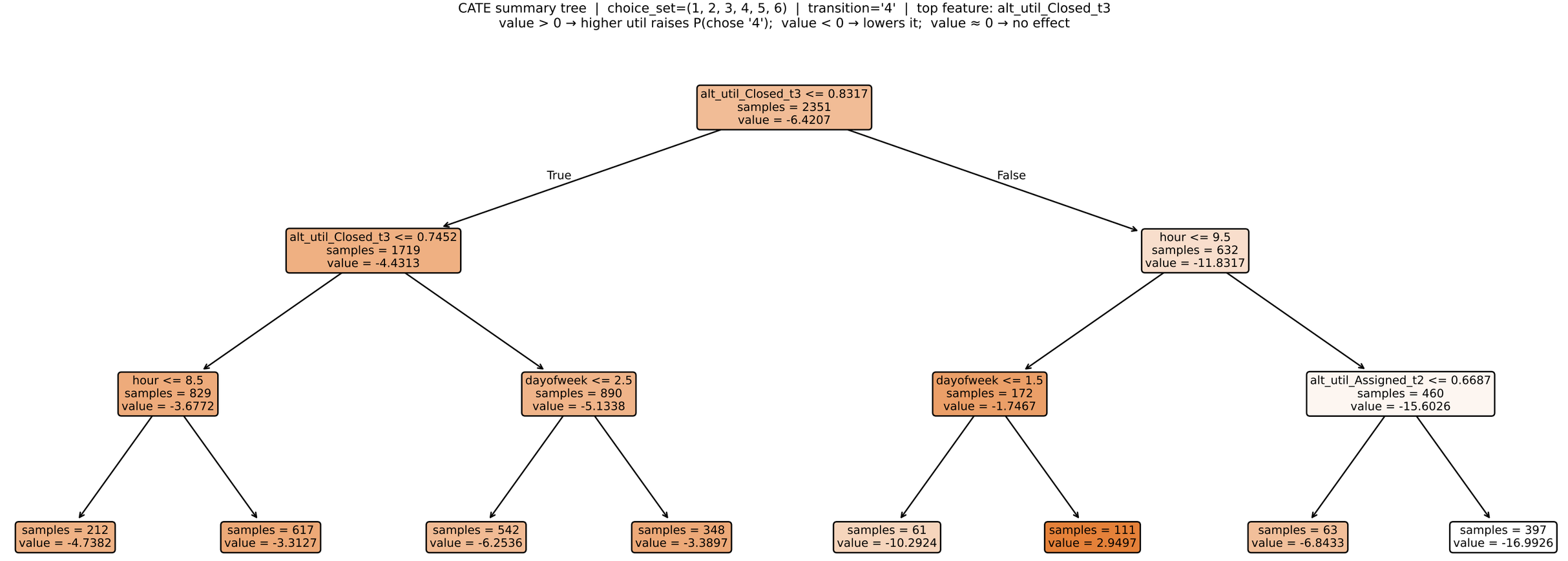

CATE — choice_set=(1, 2, 3, 4, 5, 6) | transition='2' | n=2351 | ATE=-1.7660 std=1.6208
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.7660
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

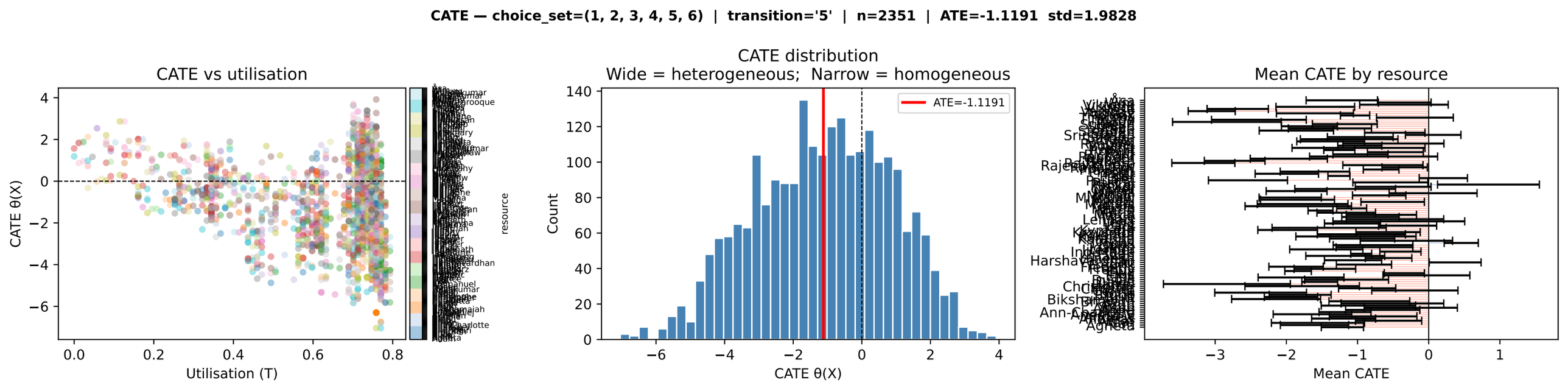

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6) | transition='2' | top feature: alt_util_Wait_t5
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Wait_t5 <= 0.7644
samples = 2351
value = -1.766
True
False
alt_util_Awaiting Assignment_t1 <= 0.8397
samples = 2100
value = -2.0678
dayofweek <= 1.5
samples = 251
value = 0.7588
alt_util_Wait_t5 <= 0.7184
samples = 1766
value = -2.2579
hour <= 10.5
samples = 334
value = -1.0628
hour <= 11.5
samples = 99
value = -0.3823
hour <= 12.5
samples = 152
value = 1.502
samples = 1058
value = -1.8763
samples = 708
value = -2.8281
samples = 133
value = 0.0684
samples = 201
value = -1.8112
samples = 47
value = 0.2339
samples = 52
value = -0.9393
samples = 98
value = 2.0499
samples = 54
value = 0.5077

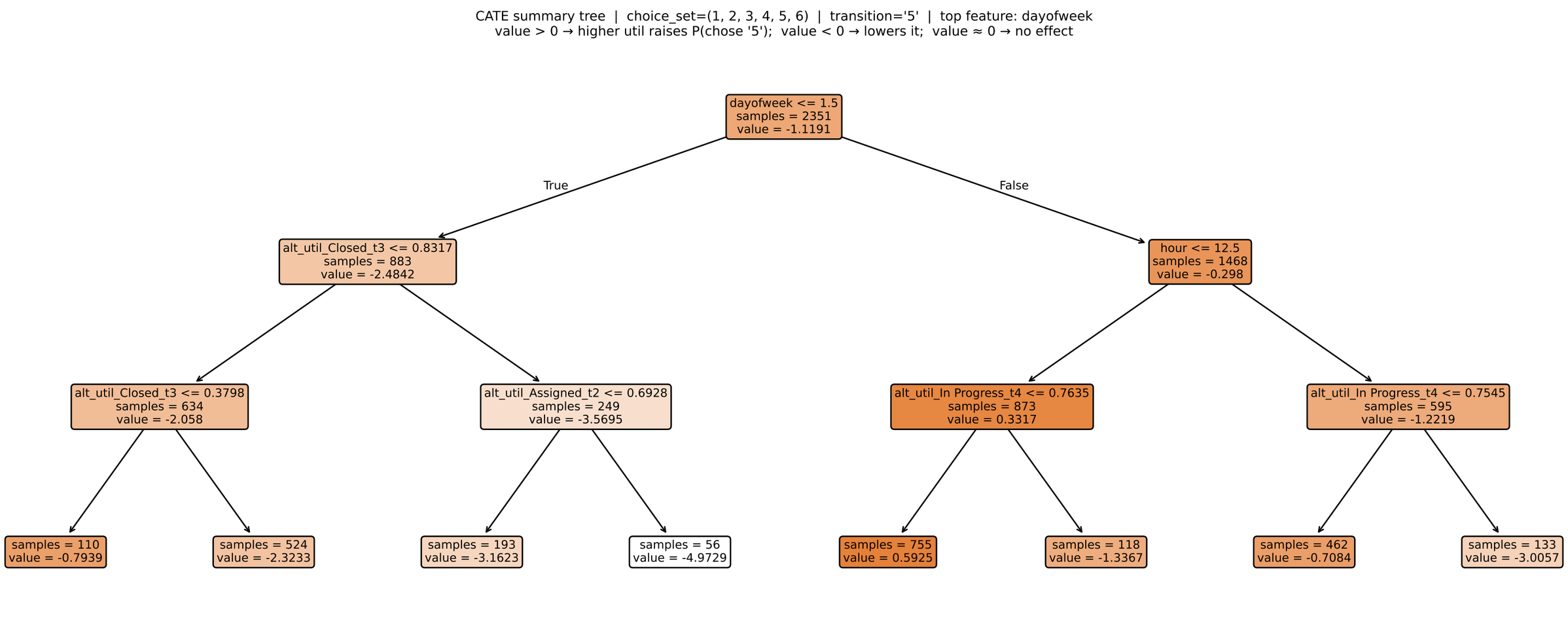

CATE — choice_set=(1, 2, 3, 4, 5, 6) | transition='3' | n=2351 | ATE=-2.3232 std=2.6672
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-2.3232
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6) | transition='3' | top feature: alt_util_In Progress_t4
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 2.5
samples = 2351
value = -2.3232
True
False
alt_util_Wait_t5 <= 0.7299
samples = 1432
value = -3.2841
alt_util_In Progress_t4 <= 0.768
samples = 919
value = -0.8258
alt_util_In Progress_t4 <= 0.7365
samples = 784
value = -2.576
hour <= 13.5
samples = 648
value = -4.1408
alt_util_In Progress_t4 <= 0.7455
samples = 851
value = -0.4868
alt_util_Assigned_t2 <= 0.6807
samples = 68
value = -5.0683
samples = 669
value = -2.8294
samples = 115
value = -1.1017
samples = 460
value = -4.8545
samples = 188
value = -2.3944
samples = 642
value = -1.3118
samples = 209
value = 2.0472
samples = 9
value = 0.2639
samples = 59
value = -5.8817


CATE — choice_set=(1, 2, 3, 4, 5, 6) | transition='4' | n=2351 | ATE=-6.4207 std=6.5843
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-6.4207
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6) | transition='4' | top feature: alt_util_Closed_t3
value > 0 → higher util raises P(chose '4'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_Closed_t3 <= 0.8317
samples = 2351
value = -6.4207
True
False
alt_util_Closed_t3 <= 0.7452
samples = 1719
value = -4.4313
hour <= 9.5
samples = 632
value = -11.8317
hour <= 8.5
samples = 829
value = -3.6772
dayofweek <= 2.5
samples = 890
value = -5.1338
dayofweek <= 1.5
samples = 172
value = -1.7467
alt_util_Assigned_t2 <= 0.6687
samples = 460
value = -15.6026
samples = 212
value = -4.7382
samples = 617
value = -3.3127
samples = 542
value = -6.2536
samples = 348
value = -3.3897
samples = 61
value = -10.2924
samples = 111
value = 2.9497
samples = 63
value = -6.8433
samples = 397
value = -16.9926


CATE — choice_set=(1, 2, 3, 4, 5, 6) | transition='5' | n=2351 | ATE=-1.1191 std=1.9828
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-1.1191
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(1, 2, 3, 4, 5, 6) | transition='5' | top feature: dayofweek
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 1.5
samples = 2351
value = -1.1191
True
False
alt_util_Closed_t3 <= 0.8317
samples = 883
value = -2.4842
hour <= 12.5
samples = 1468
value = -0.298
alt_util_Closed_t3 <= 0.3798
samples = 634
value = -2.058
alt_util_Assigned_t2 <= 0.6928
samples = 249
value = -3.5695
alt_util_In Progress_t4 <= 0.7635
samples = 873
value = 0.3317
alt_util_In Progress_t4 <= 0.7545
samples = 595
value = -1.2219
samples = 110
value = -0.7939
samples = 524
value = -2.3233
samples = 193
value = -3.1623
samples = 56
value = -4.9729
samples = 755
value = 0.5925
samples = 118
value = -1.3367
samples = 462
value = -0.7084
samples = 133
value = -3.0057

| | | |

CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(1, 2, 3, 4, 5, 6)	1	2351	-2.894506	1.37114	-7.342006	-3.857851	-2.820038	-1.818257	0.729942	0.7	99.3	alt_util_In Progress_t4
(1, 2, 3, 4, 5, 6)	2	2351	-1.766046	1.620813	-5.866871	-2.773303	-1.892276	-0.870369	5.518349	12.4	87.6	alt_util_Wait_t5
(1, 2, 3, 4, 5, 6)	3	2351	-2.323152	2.667234	-11.711586	-3.810251	-2.148221	-0.965396	11.076199	10.5	89.5	alt_util_In Progress_t4
(1, 2, 3, 4, 5, 6)	4	2351	-6.420711	6.584349	-33.801449	-8.524956	-4.772925	-2.670442	12.621271	7.5	92.5	alt_util_Closed_t3
(1, 2, 3, 4, 5, 6)	5	2351	-1.119095	1.982824	-7.041056	-2.588484	-1.024414	0.40236	3.917382	32.1	67.9	dayofweek

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md